

Assignment 2

1 Problem 1

1.1 Problem 1.1

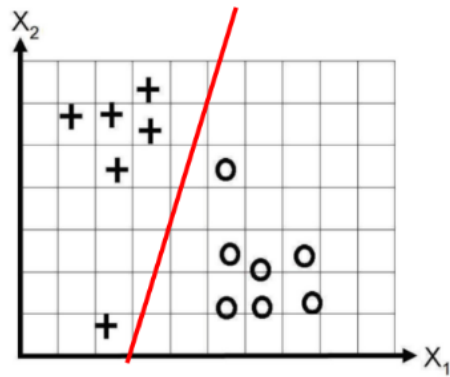


Figure 1: Visualization of data

The decision boundary is not unique. The classification error is 0.

1.2 Problem 1.2

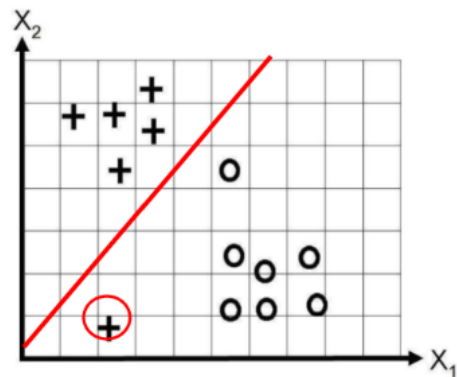


Figure 2: Visualization of data

The decision boundary is unique and must cross the original point $(0, 0)$. The classification error is 1.

1.3 Problem 1.3

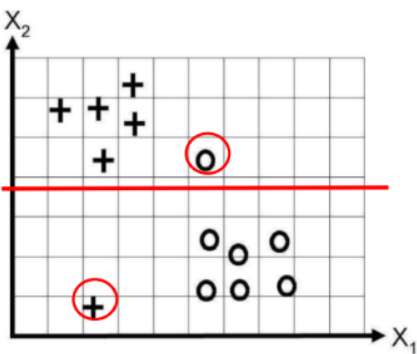


Figure 3: Visualization of data

The decision boundary is a horizontal line and is unique. The classification error is 2.

1.4 Problem 1.4

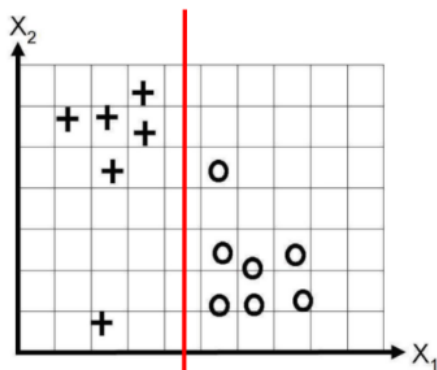


Figure 4: Visualization of data

The decision boundary is a vertical line and is unique. The classification error is 0.

2 Problem 2

2.1 Problem 2.1

Since

$$\phi(\mathbf{x}_1) = [1, 0, 0]^T, \phi(\mathbf{x}_2) = [1, 2, 2]^T$$

So the vector parallel to the optimal value can be calculate as:

$$\phi(\mathbf{x}_2) - \phi(\mathbf{x}_1) = [0, 2, 2]^T$$

2.2 Problem 2.2

Because we have 2 points, the margin should be half of the distance between $\phi(\mathbf{x}_1)$ and $\phi(\mathbf{x}_2)$:

$$d = \sqrt{0 + 4 + 4} = 2\sqrt{2}$$
$$\text{margin} = \frac{1}{2} * d = \sqrt{2}$$

2.3 Problem 2.3

$$\sqrt{2} = \frac{1}{\|\mathbf{w}\|}, \|\mathbf{w}\| = \frac{\sqrt{2}}{2}$$

by 1.2.1, assume $w = [0, a, a]^T$, we can solve $|a| = \frac{1}{2}$

Since:

$$y_1(\mathbf{w}^T \phi(x_1) + w_0) \geq 1$$

$$y_2(\mathbf{w}^T \phi(x_2) + w_0) \geq 1$$

we will get:

$$-w_0 \geq 1$$

$$4a + w_0 \geq 1$$

so $a = \frac{1}{2}$, $\mathbf{w} = [0, \frac{1}{2}, \frac{1}{2}]^T$ You could also check the dual feasibility to get the same result.

2.4 Problem 2.4

Tight the inequalities in 1.2.3, we will get $w_0 = -1$

2.5 Problem 2.5

$$f(x) = \mathbf{w}^T \phi(x) + w_0 = \frac{1}{2}x^2 + \frac{\sqrt{2}}{2}x - 1$$

3 Problem 3

3.1 Problem 3.1

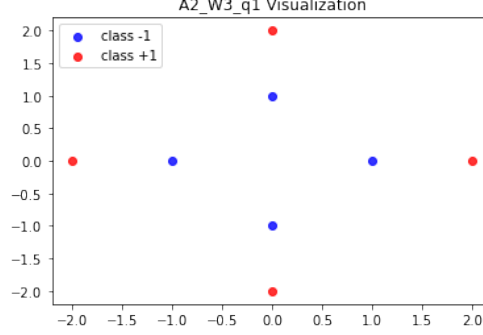


Figure 5: Visualization of data

Since the data is not linearly separable, we cannot find an SVM classifier without slack variable for this dataset.

(It also makes sense to say that it's possible to find one when we are able to add a suitable nonlinear kernel)

3.2 Problem 3.2

After expanding the original feature vector, we get the new labels:

$$\text{Class -1: } \begin{bmatrix} (1 & 0) \\ (0 & 1) \\ (1 & 0) \\ (0 & 1) \end{bmatrix} \quad \text{Class +1: } \begin{bmatrix} (4 & 0) \\ (0 & 4) \\ (4 & 0) \\ (0 & 4) \end{bmatrix}$$

By plotting the data, we can easily find that all the data are support vectors for the SVM. And the margin of the SVM is $\frac{3\sqrt{2}}{4}$ (Half of the distance between the two black lines).

We also see that $\mathbf{w}$ is orthogonal to the line: $x_2 + x_1 = 1$. Thus $\mathbf{w}^T(1, -1) = 0$. This shows that $w_1 = w_2$.

Thus, $\text{margin} = \frac{1}{\|\mathbf{w}\|} = \frac{1}{\sqrt{w_1^2 + w_2^2}} = \frac{3\sqrt{2}}{4}$. We get $w_1 = w_2 = \frac{2}{3}$.

By plugging in one point $(1, 0)$ (on the boundary) to the constraint, we get $y_1(\mathbf{w}^T\phi(x_1) + w_0) = 1$, $w_0 = -\frac{5}{3}$

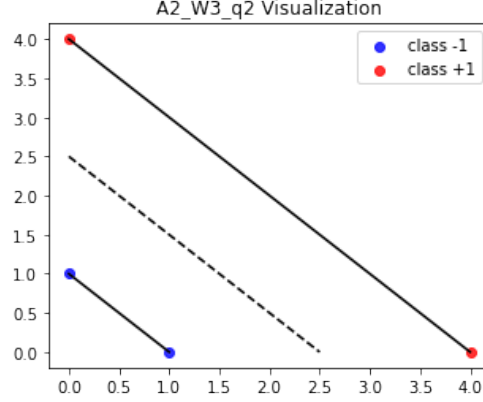


Figure 6: Visualization of expanded data

Thus the decision boundary for the SVM is $f(\phi(\mathbf{x})) = \left(\frac{2}{3}, \frac{2}{3}\right)^T \phi(\mathbf{x}) - \frac{5}{3}$

We will predict class +1 when $f(\phi(\mathbf{x})) > 0$ and class -1 otherwise.

Predicting for (1, 2): $f(\phi((1, 2))) = \left(\frac{2}{3}, \frac{2}{3}\right)^T \phi((1, 2)) - \frac{5}{3} = \frac{2}{3} + \frac{8}{3} - \frac{5}{3} = \frac{5}{3} > 0$. Thus (1, 2) belongs to class +1.

4 Problem 4

The objective function of the support vector machine is:

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

subject to $y_i(w^T x_i + b) \geq 1, \forall i$

Then, the Lagrangian function becomes:

$$L(\alpha, w, b) = \frac{1}{2} \|w\|^2 - \sum_{n=1}^N \alpha_n [y_n(w^T x_n + b) - 1]$$

Since the second term (including the negative sign) is always negative, the Lagrange primal function p is:

$$p(\alpha) = \frac{1}{2} \|w\|^2$$

Then, the primal problem becomes:

$$\min_{\alpha \geq 0} \frac{1}{2} \|w\|^2$$

Next, the Lagrange dual function d is:

$$d(\alpha) = \min L(\alpha, w, b)$$

By KKT conditions:

$$\begin{aligned}\frac{\partial L}{\partial w} &= w - \sum_{n=1}^N \alpha_n y_n x_n = 0 \\ \Rightarrow w &= \sum_{n=1}^N \alpha_n y_n x_n\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial b} &= - \sum_{n=1}^N \alpha_n y_n = 0 \\ \Rightarrow \sum_{n=1}^N \alpha_n y_n &= 0\end{aligned}$$

Then, we get

$$\begin{aligned}- \sum_{n=1}^N \alpha_n [y_n (w^T x_n + b) - 1] &= - \sum_{n=1}^N \alpha_n y_n \left(\sum_{m=1}^N \alpha_m y_m x_m \right)^T x_n + \sum_{n=1}^N \alpha_n \\ &= - \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m x_n^T x_m + \sum_{n=1}^N \alpha_n\end{aligned}$$

Therefore, the dual function becomes:

$$\begin{aligned}d(\alpha) &= \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m y_n y_m x_n^T x_m \\ &= \sum_{n=1}^N \alpha_n - \frac{1}{2} \|w\|^2 \\ \text{s.t. } \sum_{n=1}^N \alpha_n y_n &= 0\end{aligned}$$

Since both primal and dual problems have the same optimal value.

$$\sum_{n=1}^n \alpha_n - \frac{1}{2} \|w\|^2 = \frac{1}{2} \|w\|^2$$

Then, by $\gamma = \|w\|$:

$$\sum_{n=1}^n \alpha_n = \|w\|^2 = \frac{1}{\gamma^2}$$